



Introduction To

ROS MoveIt



ROS

Robot Operating System





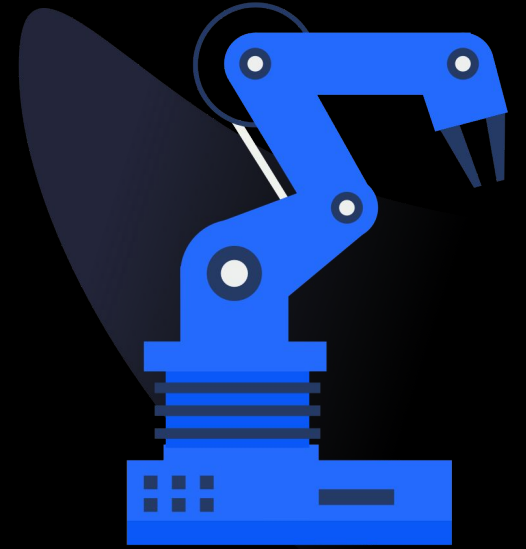
Using ROS MoveIt & Navigation Stack

What is *MoveIt*

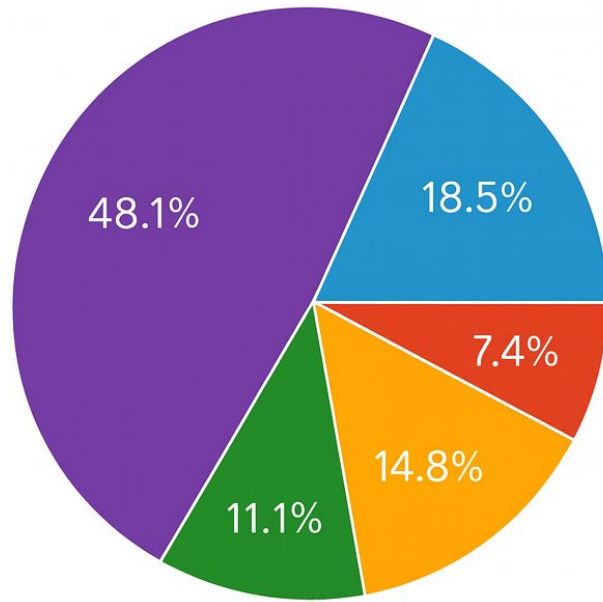
- **MoveIt** is a motion planning, manipulation, and kinematics framework for ROS, ideal for students and university researchers.
- **MoveIt** is an **open-source project** that is the result of the combined efforts of a large international community and multiple organizations.
- **MoveIt** has been used on over **150+ robots** by the community.

Why MoveIt?

- MoveIt is the most widely used software for manipulation and has been used on over 150 robots.
- It is released under the terms of the BSD license, and thus free for industrial, commercial, and research use.
- By incorporating the latest advances in motion planning, manipulation, 3D perception, kinematics, control and navigation, MoveIt is state of the art software for mobile manipulation.



Movelt Survey in 2019



- Undergraduate Student
- Graduate Student
- Academic Faculty / Post-doc
- Researcher at Government Lab / Non-Profit R&D Lab
- Industry Professional at Small Company / Startup
- Hobbyist

Movelt: **Evolution** of a Motion Planning Platform

arm_navigation



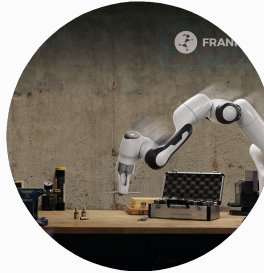
Version
1.0
201
0

Movelt!



Beta
201
3

Movelt!



Version
1.0
201
9

Movelt2



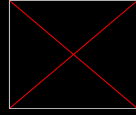
Version
2.0
202
0

MoveIt 1 vs MoveIt 2

MoveIt 1	MoveIt 2
Runs on ROS 1	Runs on ROS 2
Limited real-time support	Better real-time behavior via ROS 2 (DDS, QoS)
Older, less modular architecture	Component-based , more modular and scalable
Mostly single-threaded in practice	Designed for multi-threaded use with component nodes
Uses older ROS control stack	Integrates with ROS 2 Control for cleaner hardware interfaces
Limited support for parallel planning	Supports parallel planning pipelines (multiple planners in parallel)
Mature but based on legacy ROS 1 ecosystem	Actively developed, future-focused in ROS 2 ecosystem
Good for existing ROS 1 deployments	Recommended for new projects and modern applications

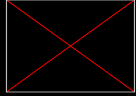
Functionalities of MoveIt

1. Motion Planning



- Generate high-degree of freedom trajectories through cluttered environments and avoid local minimums

2. Manipulation



- Analyze and interact with your environment with grasp generation

Functionalities of MoveIt

3. Inverse Kinematics

- Solve for joint positions for a given pose, even in over-actuated arms

4. Control

- Execute time-parameterized joint trajectories to low level hardware controllers through common interfaces

Functionalities of MoveIt

5. 3D Perception

- Connect to depth sensors and point clouds with Octomaps

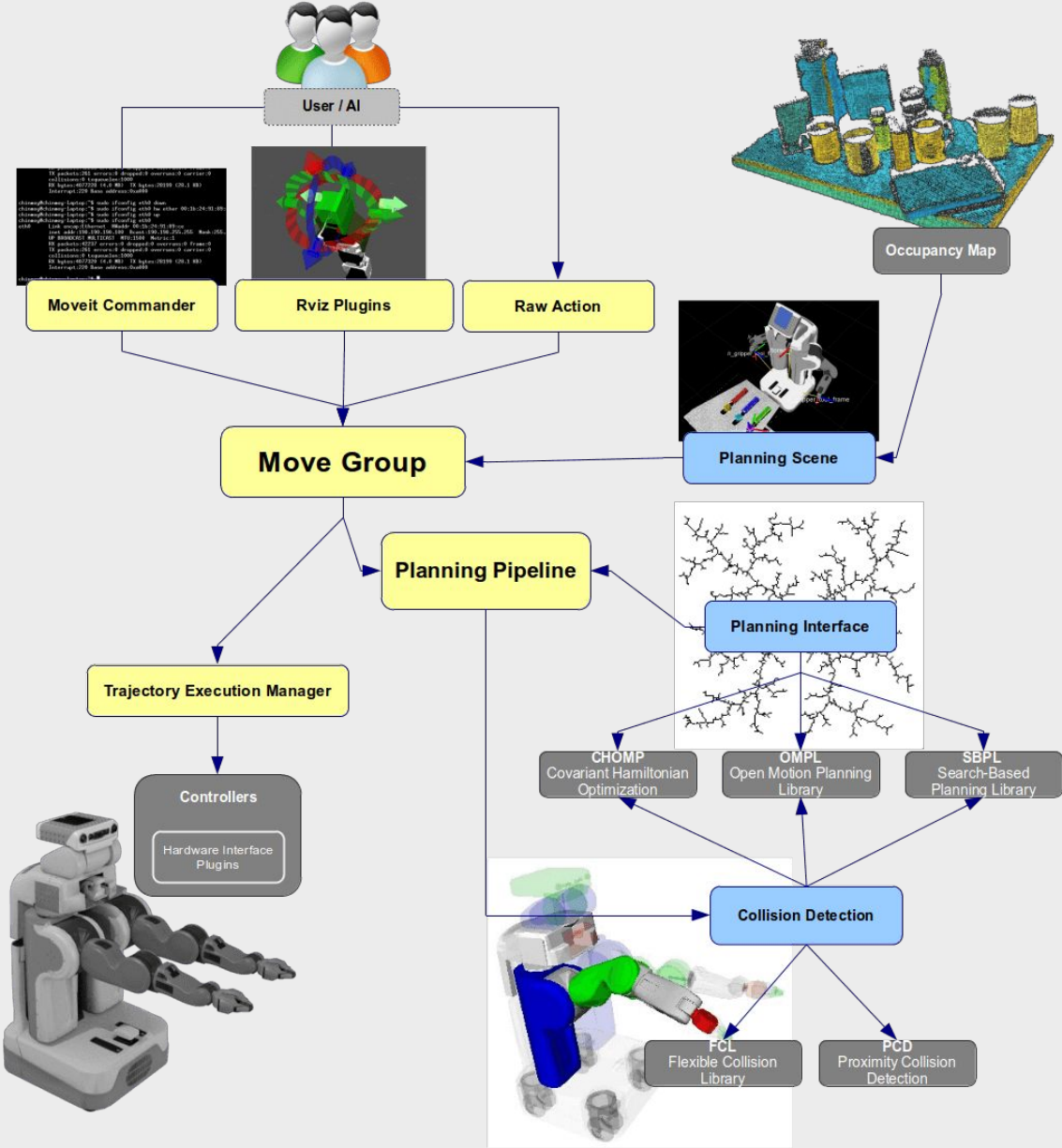
6. Collision Checking

- Avoid obstacles using geometric primitives, meshes, or point cloud data

What are the Research Libraries in MoveIt?

1. **OMPL** – Open Motion Planning Library for sampling-based motion planning.
2. **SBPL** – Search-Based Planning Library for grid and graph search planners.
3. **CHOMP** – Covariant Hamiltonian Optimization-based trajectory optimizer.
4. **STOMP** – Stochastic Trajectory Optimization for motion planning.
5. **FCL** – Fast Collision Checking Library for collision detection.
6. **PCD** – Proximity Collision Detection library for distance and collision queries.
7. **IKFast** – Analytical inverse kinematics solver that generates closed-form IK.
8. **KDL** – Kinematics and Dynamics Library for numerical IK and robot kinematics.
9. **OctoMap** – 3D occupancy grid mapping library for representing environments.

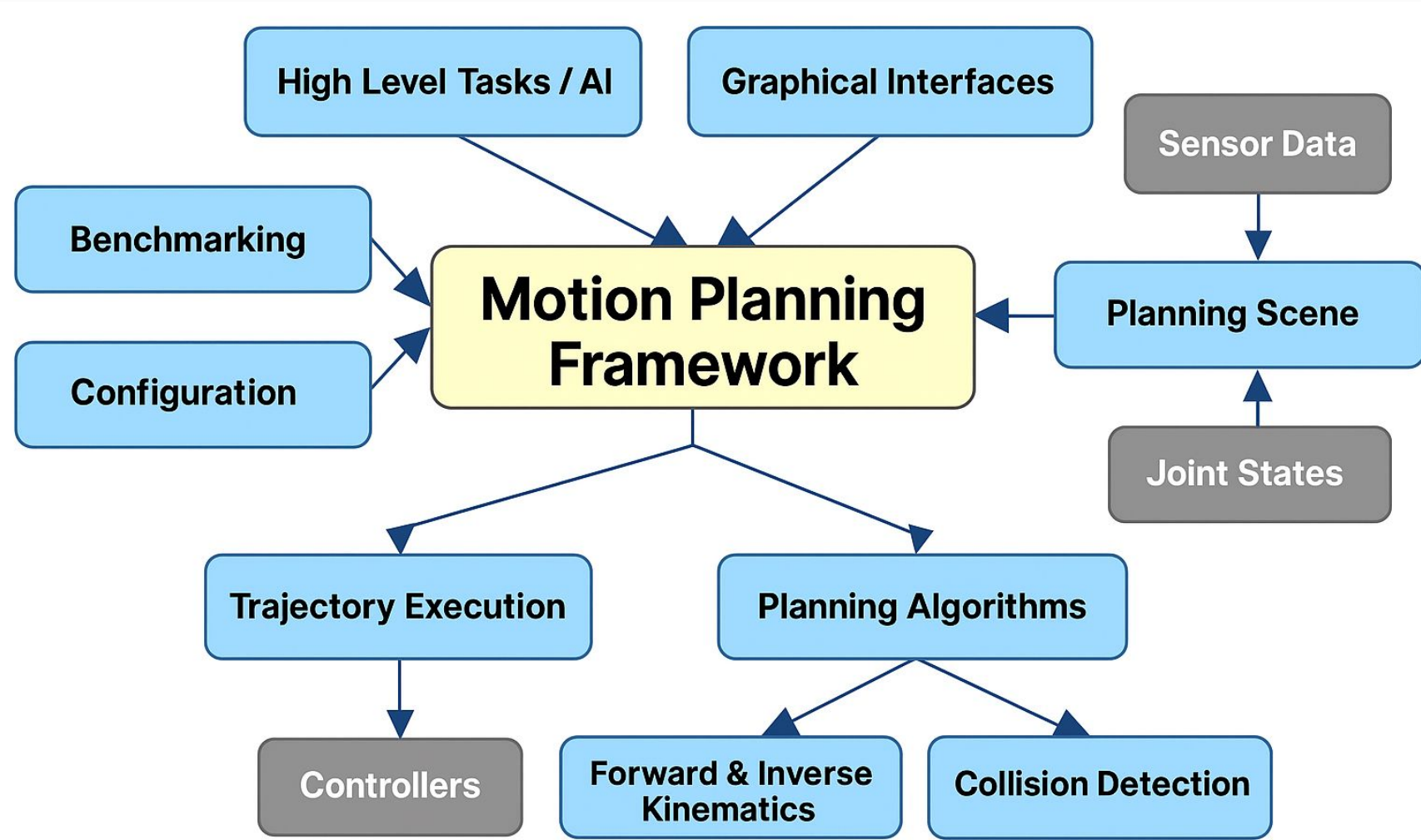
Moveit System Architecture



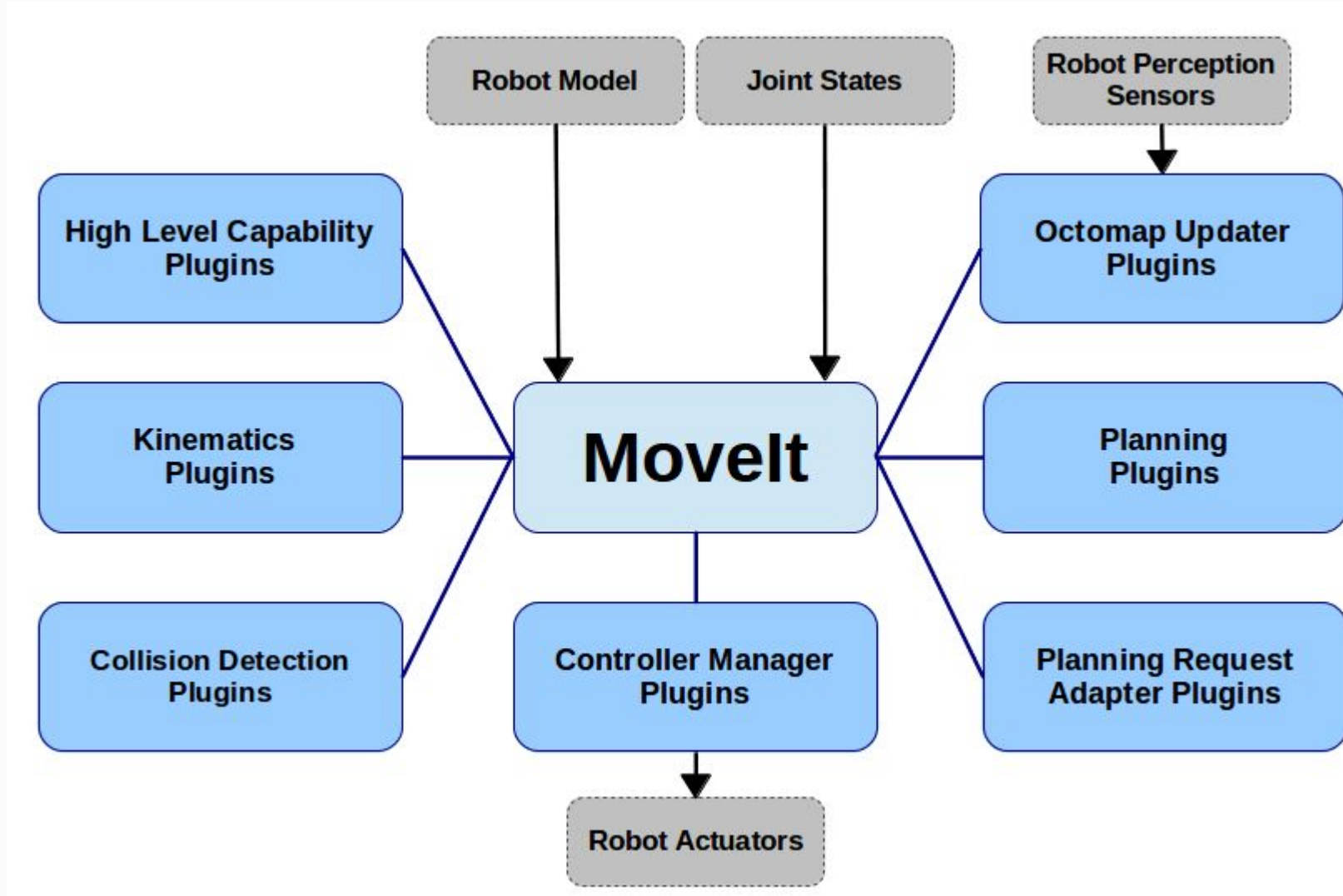
Legend



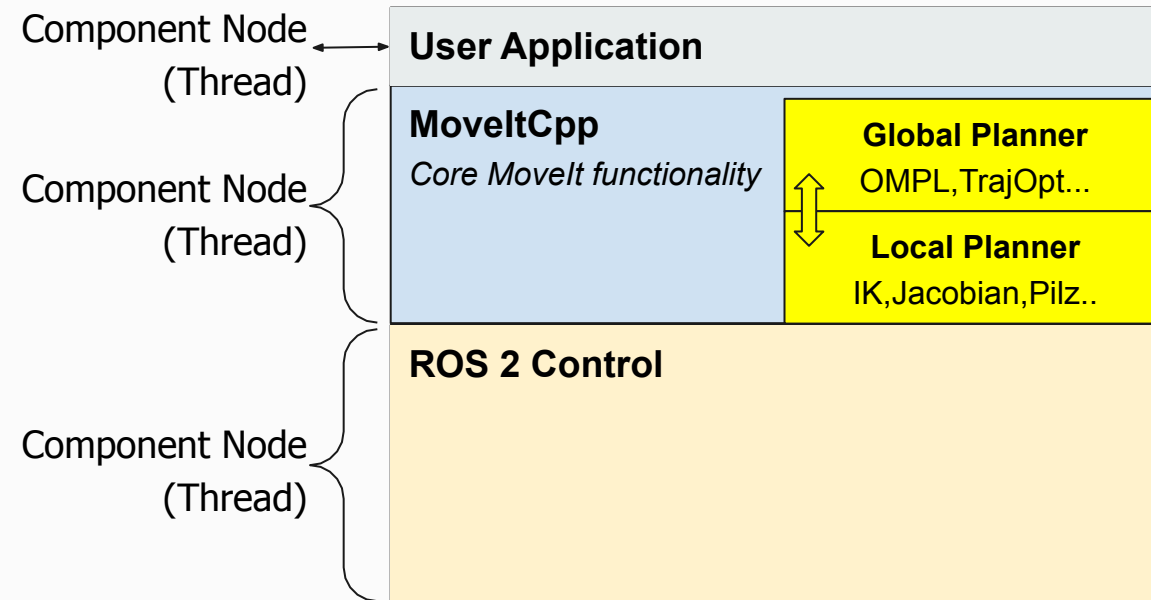
Motion Planning Framework Architecture of MoveIt



Movelt Plugin Ecosystem



Movelt Architecture with Global and Local Planners in ROS 2



Global vs Local Planners

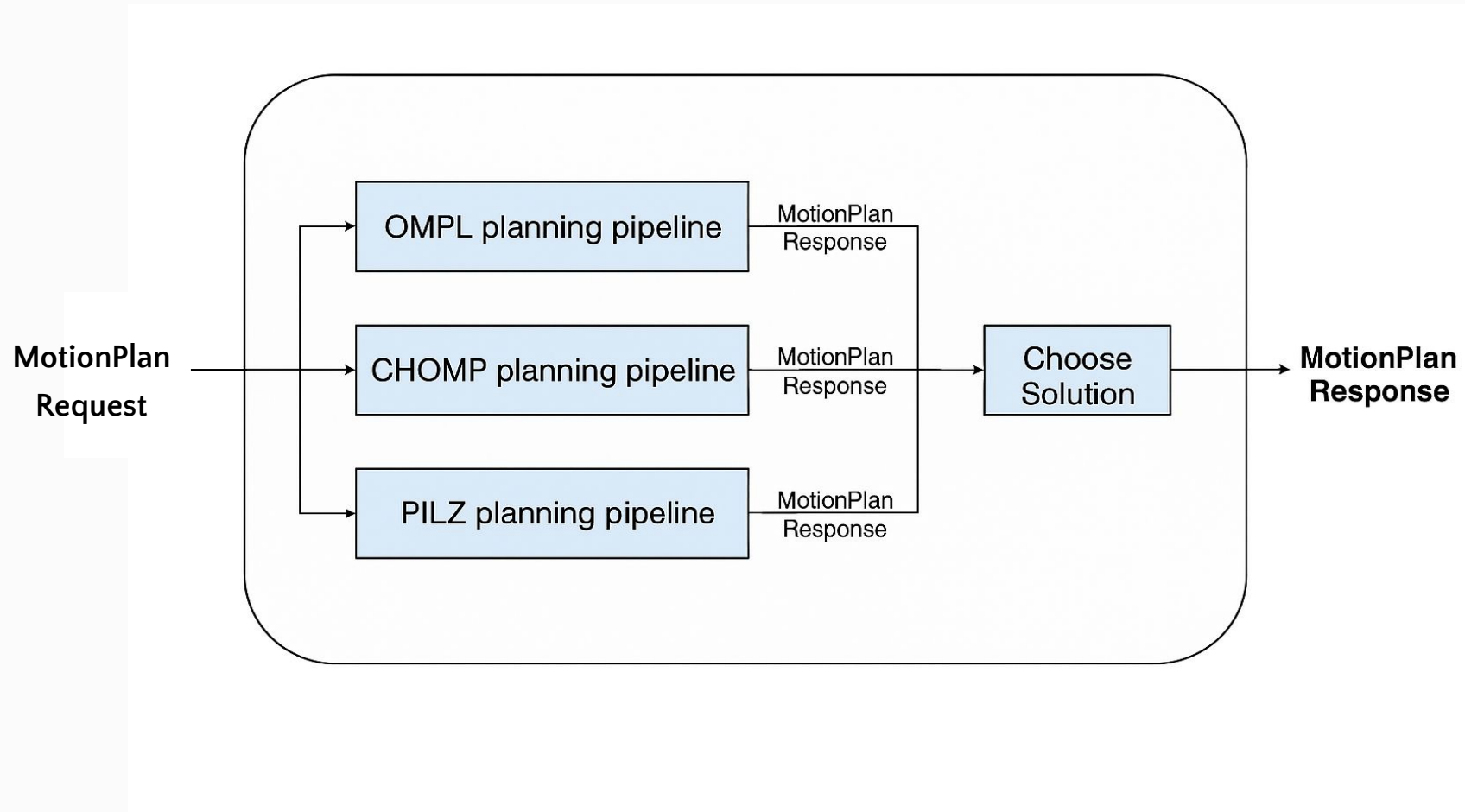
- Global planners find the **big path**; Local planners **fine-tune the small movements**.

Global Planner	Local Planner
Plans the entire path from start to goal	Makes small, short-term adjustments along the path
Works on a global map or full planning scene	Works on local, near-by robot motion
Good for long-distance, obstacle-aware planning	Good for precise movements and real-time corrections
Examples: OMPL, TrajOpt, STOMP, CHOMP	Examples: IK solvers, Jacobian-based control, Pilz planner
Slower but produces complete trajectories	Faster, used for real-time or fine control
Handles complex environments	Handles fine positioning or servoing
Used before execution	Used during execution to stay smooth and safe

How the Parallel Planning Architecture in MoveIt 2 Works

- MoveIt 2 introduces a system where **multiple motion planning algorithms run at the same time** on the same planning problem.
- This approach increases reliability and helps choose the best possible trajectory for the robot.

How the Parallel Planning Architecture in MoveIt 2 Works



How the Parallel Planning Architecture in MoveIt 2 Works

1. Starts With a MotionPlan Request

- A **MotionPlan Request** describes what the robot needs to do:
 - The start position
 - The goal position
 - ObstaclesJoint limits
 - Other planning constraints
- This **single request is sent in parallel** to several planning pipelines.

How the Parallel Planning Architecture in MoveIt 2 Works

2. Multiple Planners Run in Parallel

- Different planning pipelines try to solve the request at the same time. In the diagram, three planners are shown:
- OMPL (Open Motion Planning Library) Planning Pipeline
 - Uses sampling-based planners.
 - Good for quickly finding feasible paths.
- CHOMP (Covariant Hamiltonian Optimization for Motion Planning) Planning Pipeline
 - Uses optimization-based methods.
 - Produces smooth and refined trajectories.

How the Parallel Planning Architecture in MoveIt 2 Works

2. Multiple Planners Run in Parallel

- PILZ (Named after Pilz GmbH & Co. KG) Planning Pipeline
 - Used for industrial-style motions.
 - Very predictable and deterministic.
- Each planner works independently, using its own algorithmic approach.

How the Parallel Planning Architecture in MoveIt 2 Works

3. Each Planner Produces a MotionPlan Response

- Each pipeline outputs a MotionPlan Response, which contains:
 - A trajectory
 - Whether planning succeeded
 - Planning time
 - Path quality metrics
- So, after running, we have multiple candidate solutions to the same problem.

How the Parallel Planning Architecture in MoveIt 2 Works

4. The “Choose Solution” Step Selects the Best Path

- MoveIt 2 provides a solution selection callback, which is a user-defined function.
- This function compares all the planners' responses and selects the best one according to chosen criteria, such as:
 - Shortest path
 - Smoothest trajectory
 - Fastest planning time
 - Safest (maximum clearance from obstacles)
 - Lowest overall cost

How the Parallel Planning Architecture in MoveIt 2 Works

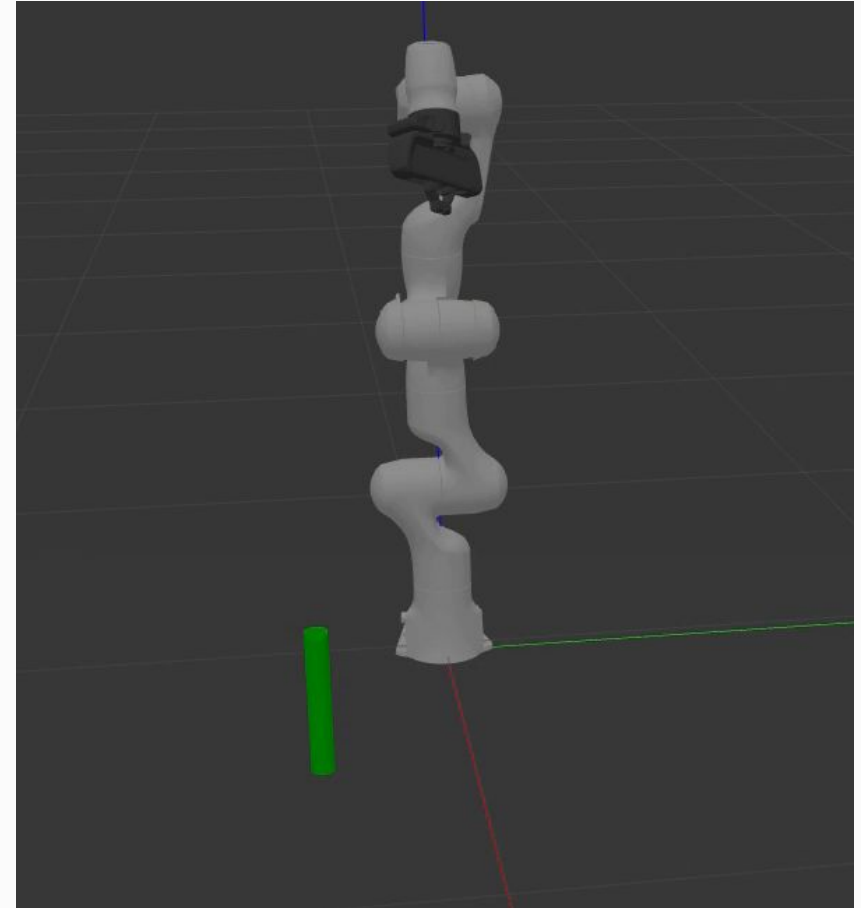
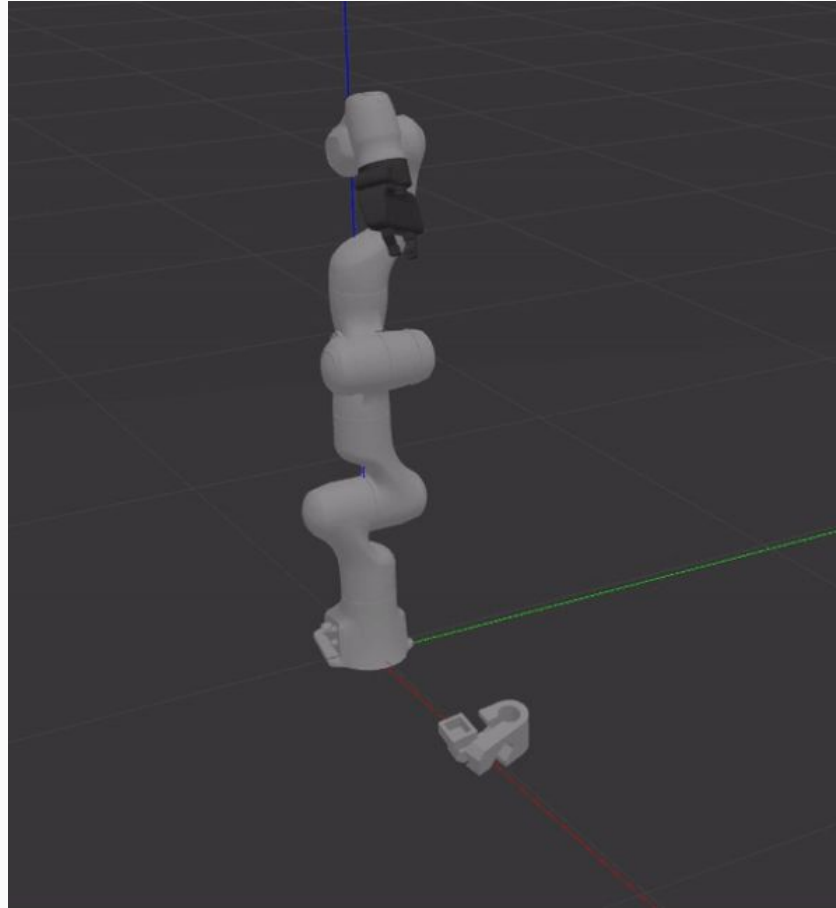
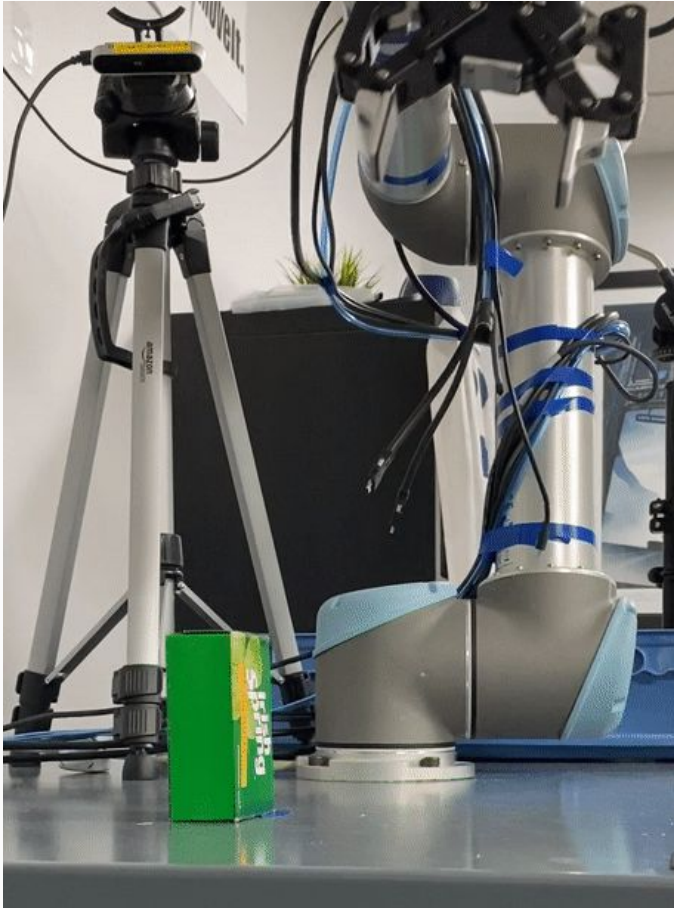
5. Final Output: A Single MotionPlan Response

- After the best trajectory is selected, it becomes the final MotionPlan Response returned to the user or to the robot's controller for execution.

Grasping using Deep Learning

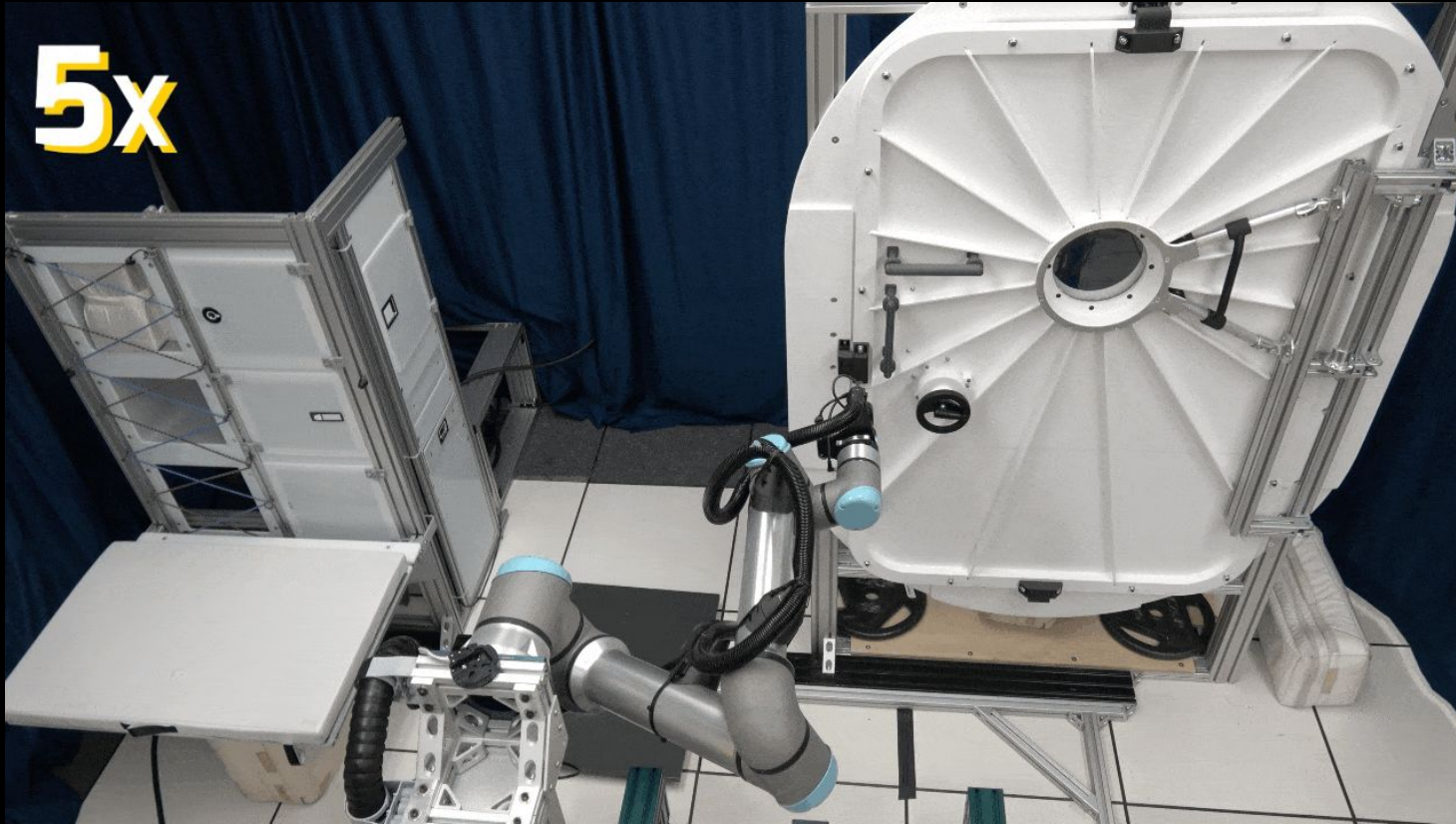
- Movelt now supports robust grasp pose generation using deep learning.
- Pick and place robots equipped with a depth camera and either a parallel jaw or suction gripper can increase productivity when paired with deep learning.
- The Movelt Task Constructor provides an interface for any grasp pose generation algorithm making Movelt's pick and place capabilities more flexible and powerful.
- Currently, the Grasp Pose Detection (GPD) library and Dex-Net are being used to detect 6-DOF grasp poses given 3D sensor data.

Grasping using Deep Learning



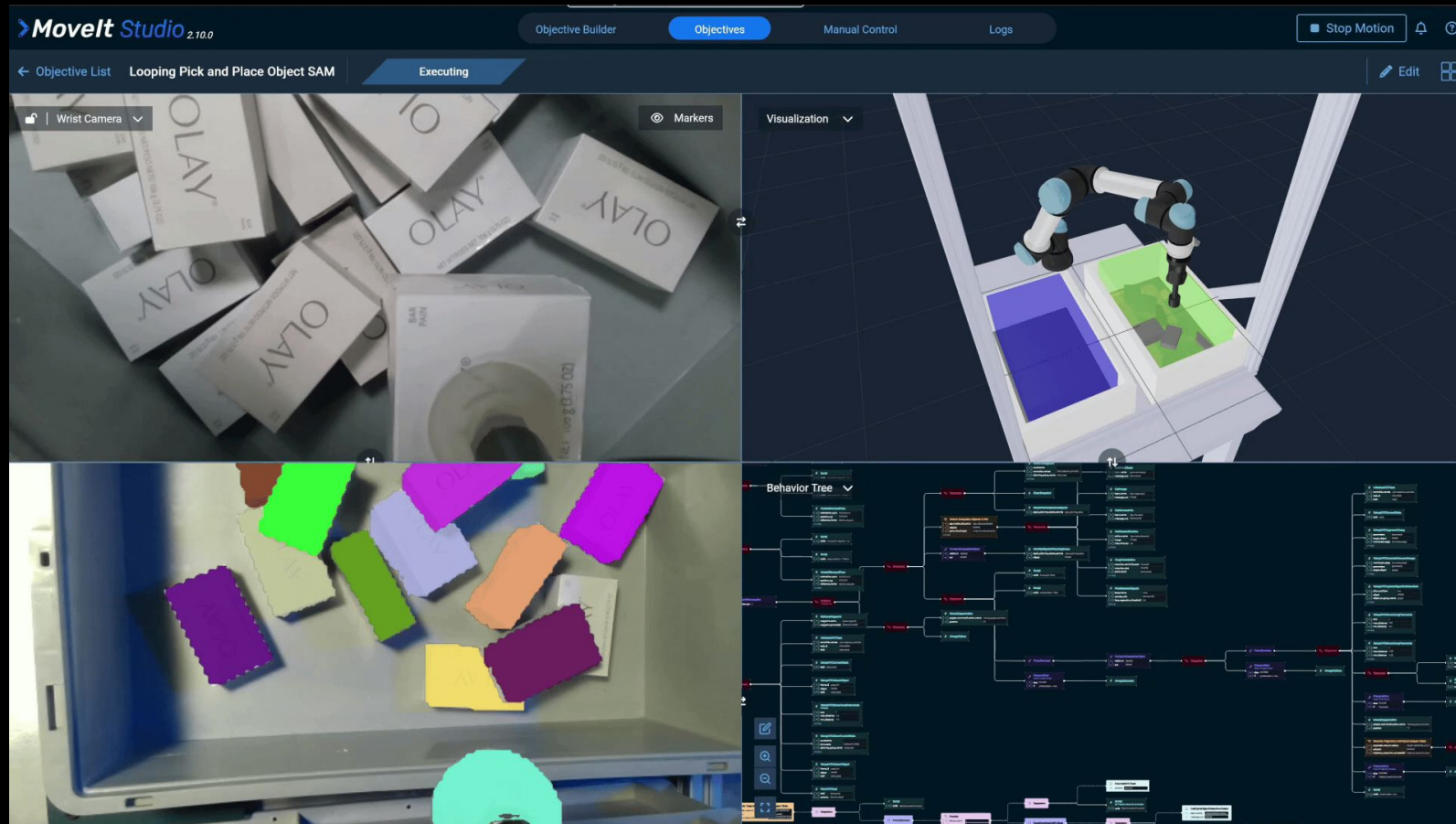
Advanced Applications of Moveit

- Complex Task Sequencing for Material Handling



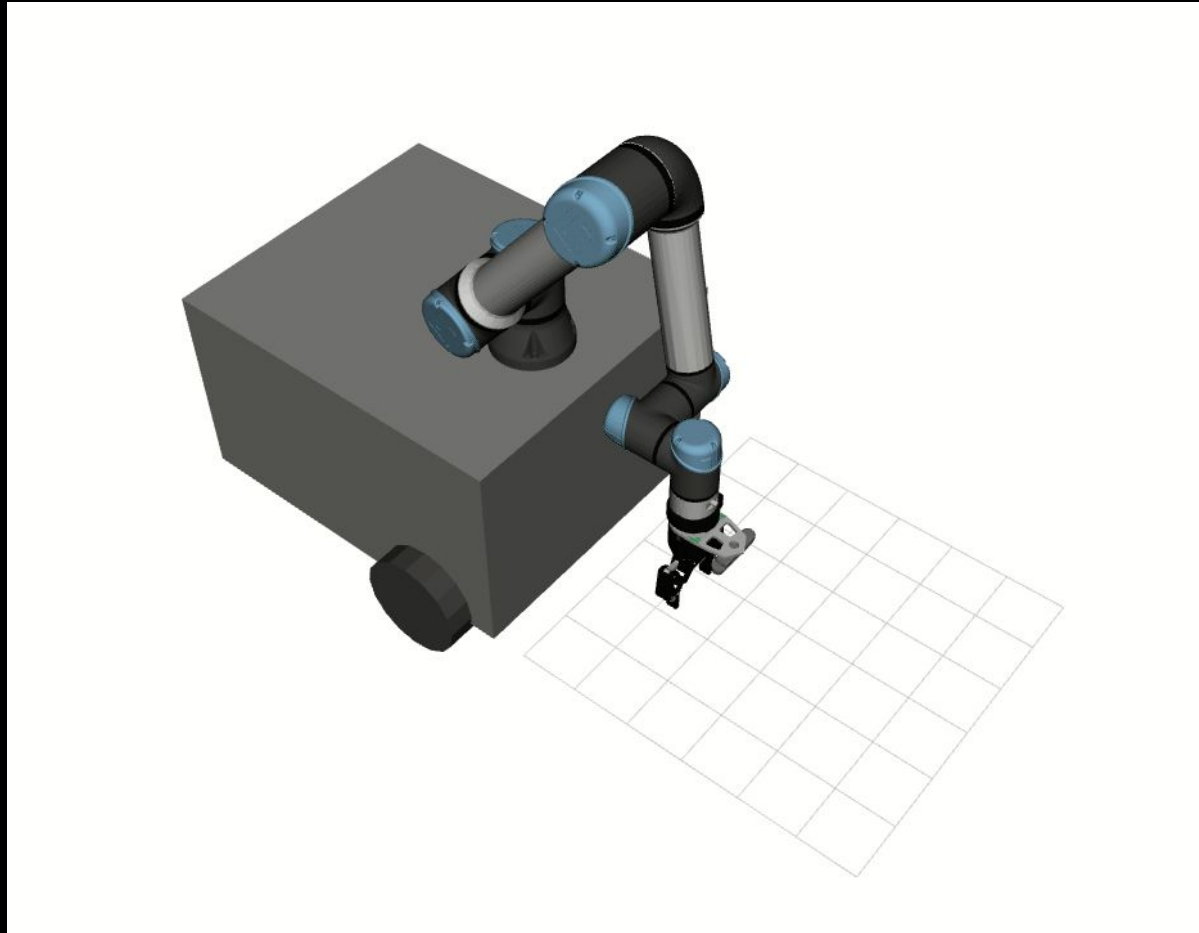
Advanced Applications of Moveit

- ML (Machine Learning)-Powered Bin Picking



Advanced Applications of Moveit

- Whole Body Cartesian Planning



Advanced Applications of Moveit

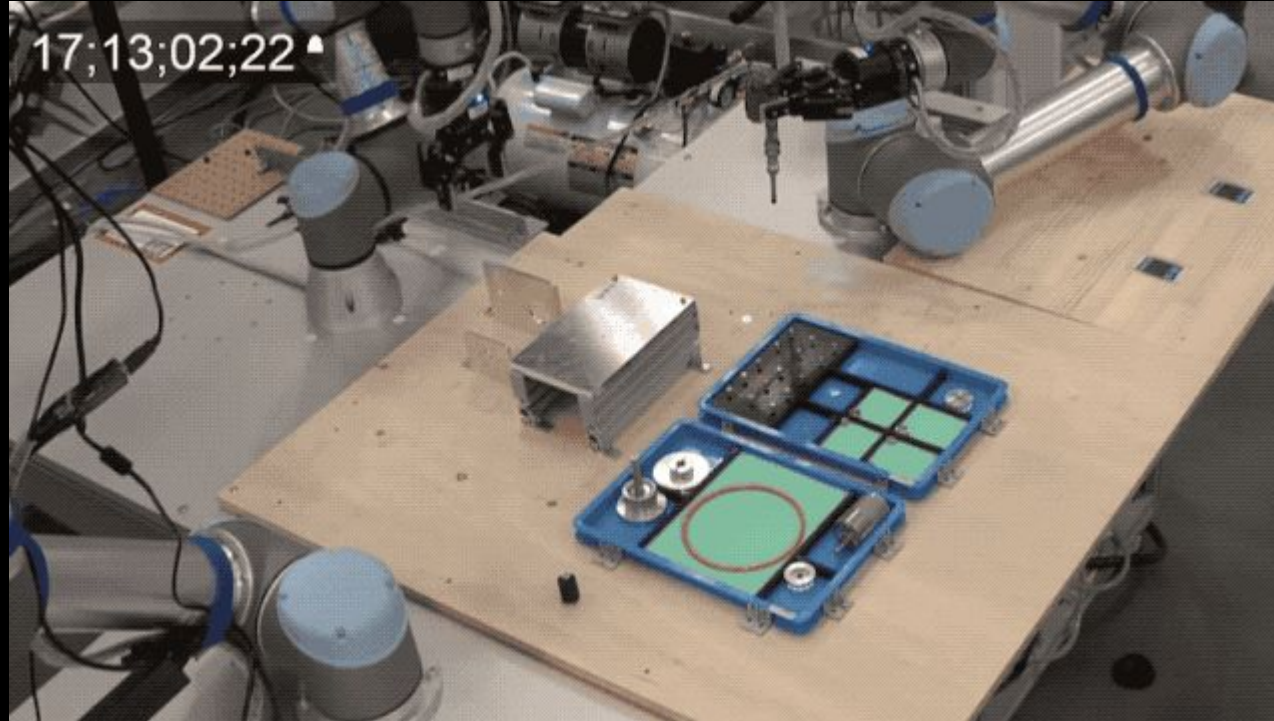
- Two robot arms, Robotic grippers
- Assembly tasks with Python examples for synchronous and simultaneous motions



Team O2AC – World Robot Summit 2020

Advanced Applications of Moveit

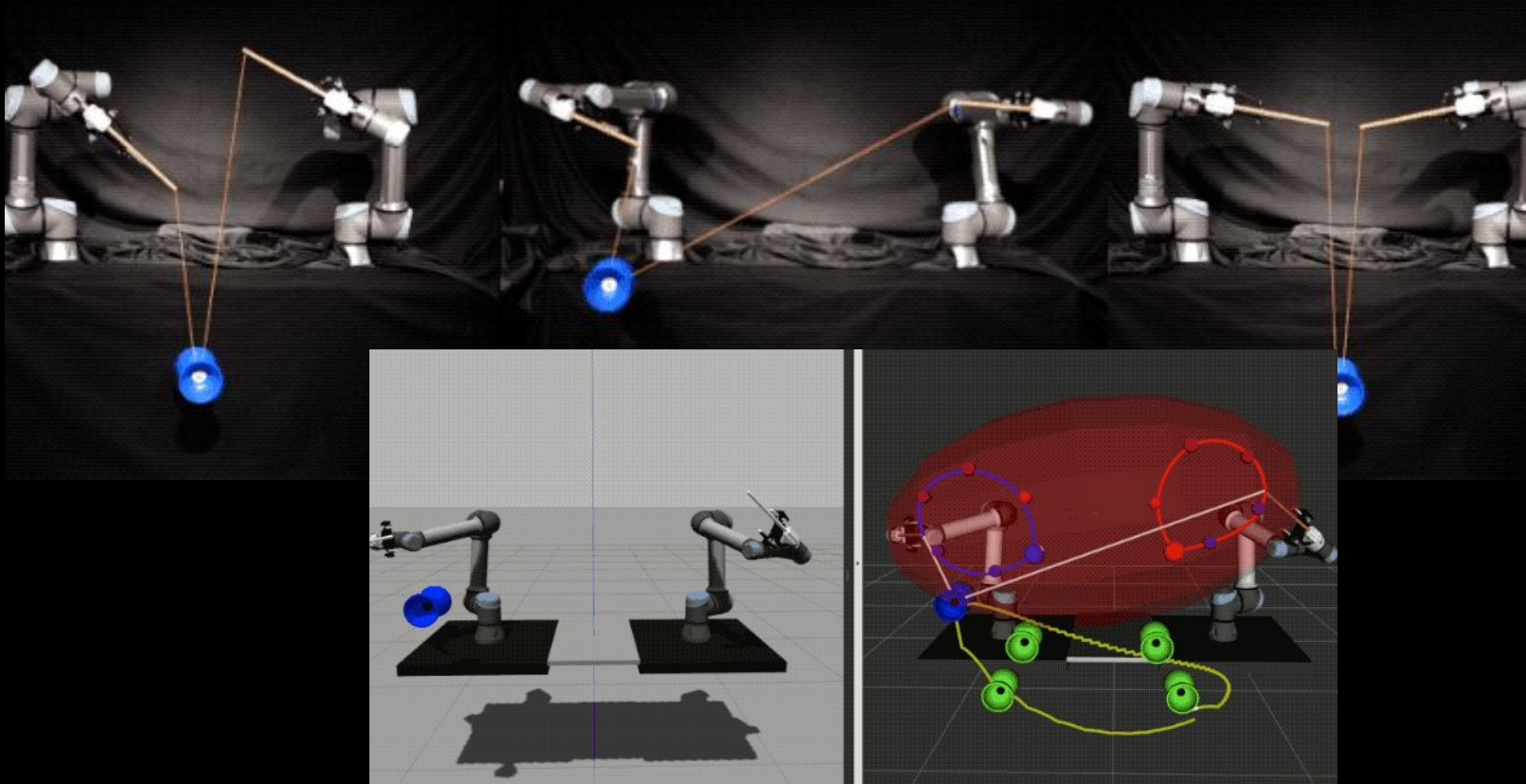
- Three-arm robot work cell
- Parametrized motions, custom assembly generator using Moveit



Team O2AS – World Robot Summit 2018

Advanced Applications of Moveit

- Two robot arms playing with a diabolo using sticks and string
- Uses Moveit and bio_ik, includes simulation and visualization



Diabolo Play (it is a juggling or circus prop consisting of an axle and two cups, played with a string attached to two hand sticks)

Land, Sea, Air with MoveIt!

Land with MoveIt!



Agriculture

Plant Harvesting



Food

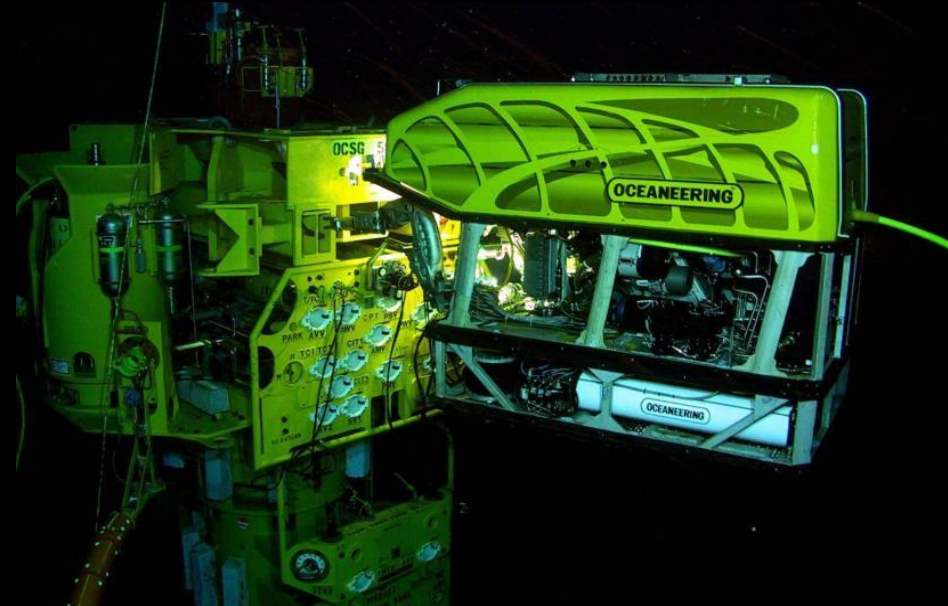
Kitchen Assistant



Logistics

Bin Picking

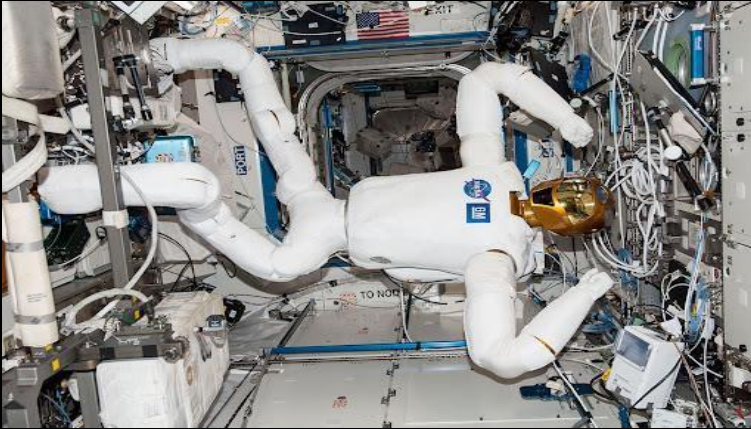
Sea with MoveIt!



Subsea

Remotely Operated
Underwater Vehicles (ROVs)

Air with MoveIt!



Space
Inter Vehicle
Robotics



Space
In Orbit Satellite
Servicing



Space
Lunar Base
Construction

ROS2 Grasp Library

Q: What is the ROS2 Grasp Library, and how does it enable deep-learning-based grasp detection and integration with MoveIt for industrial robot manipulation?

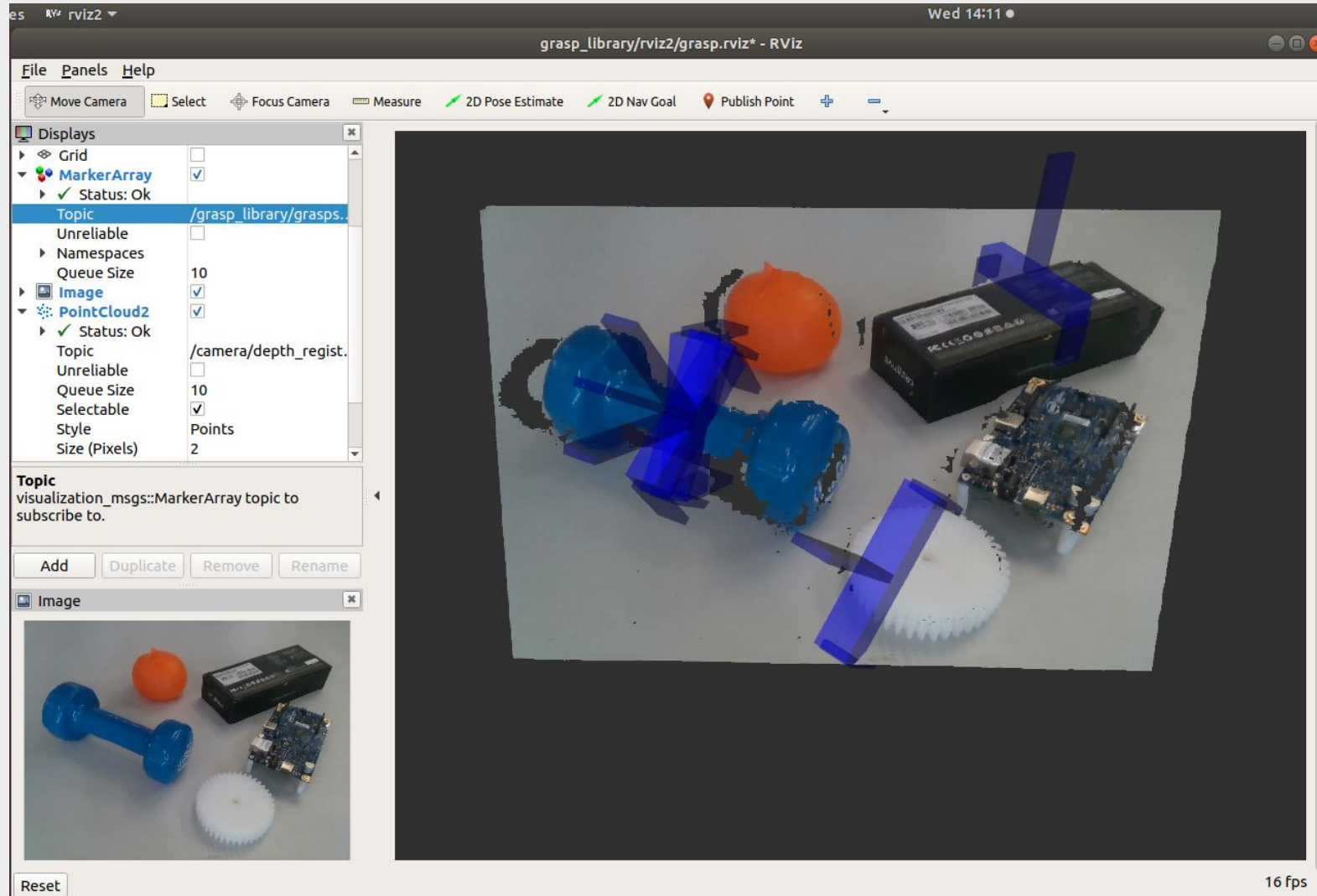
- The ROS2 Grasp Library is a ROS 2 package that brings state-of-the-art, CNN-based deep learning grasp detection to industrial robot manipulation, allowing robots to automatically identify good grasp poses from visual data such as depth images. It provides ROS 2-native interfaces that seamlessly integrate with the MoveIt motion planning framework, enabling detected grasp poses to be used directly for collision-free trajectory planning and execution. Because MoveIt and ROS-Industrial already support a wide range of industrial robot models, the ROS2 Grasp Library allows many existing robots to perform flexible, vision-guided pick-and-place tasks without the need for manually programmed grasp positions, improving adaptability, efficiency, and automation in real-world industrial environments.

Hardware Integration Challenges

Q: What is the “Chicken and Egg” problem in MoveIt’s hardware integration, and why does it slow down the adoption of ROS 2 and MoveIt 2?

- The “Chicken and Egg” problem in MoveIt happens because both users and hardware companies depend on each other to move forward, but neither wants to go first. Many users want to use ROS 2 and MoveIt 2, but they hesitate because their robot arms, sensors, or grippers don’t yet have full ROS 2 or MoveIt support. At the same time, hardware manufacturers wait for proof that many users actually need ROS 2 support before they spend time and money developing new drivers and interfaces. As a result, users don’t switch because hardware isn’t ready, and hardware doesn’t improve because users haven’t switched. This loop slows down the overall adoption of MoveIt 2 in real robots, even though the software is ready and powerful.

ROS2 Grasp Library



ROS2 Grasp Library

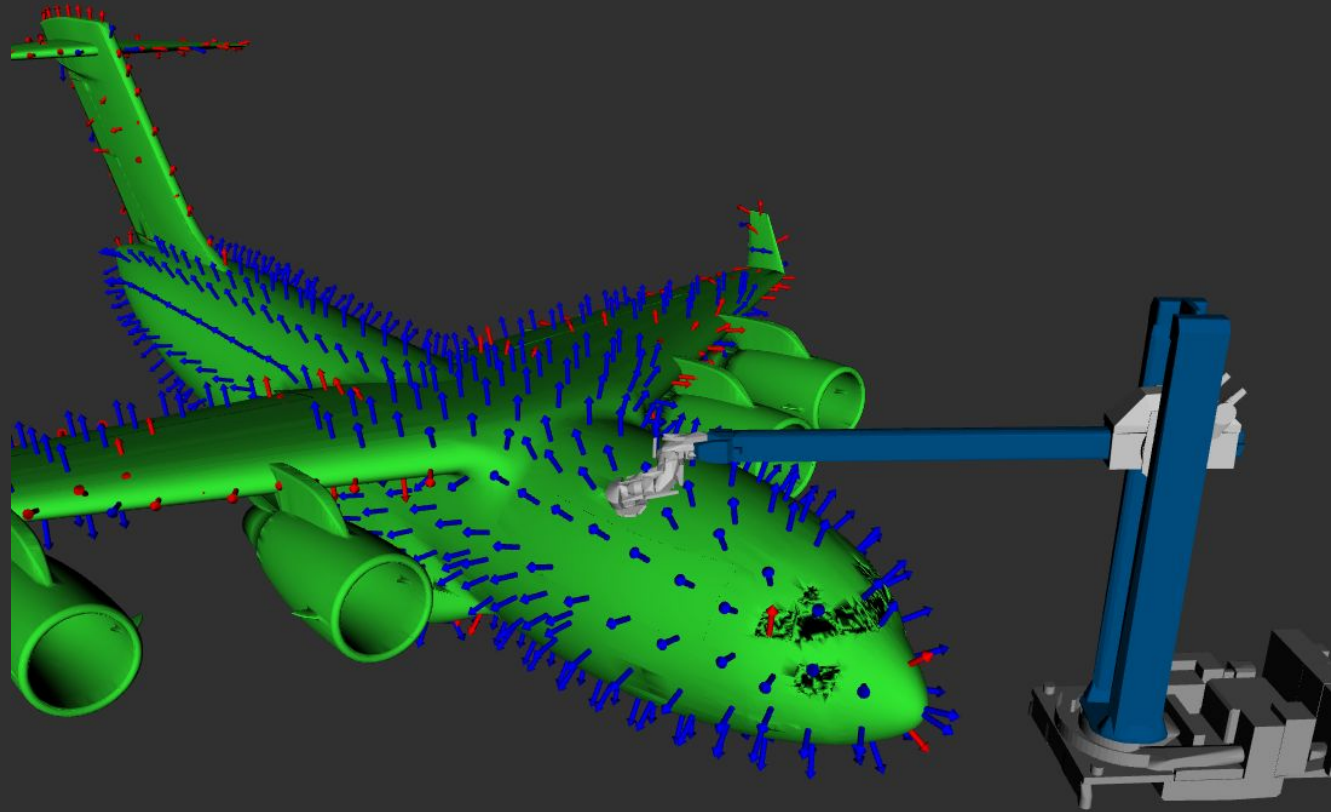
What is TracIK?

- TracIK is an **alternative Inverse Kinematics (IK) solver** designed to outperform the standard inverse Jacobian-based methods found in KDL (**Kinematics and Dynamics Library**).
- It uses smarter search and optimization methods to quickly find joint angles that put the robot's end-effector (hand/tool) at the desired position and orientation, without breaking any joint limits.
- This makes TracIK more reliable and practical for real robots, where joints cannot rotate freely forever and the robot still needs a fast, good IK solution.

What is REACH?

- REACH is a tool that helps you check which parts of a large object—like the airplane in the image—a robot can physically reach. It does this by testing many different points on the surface and marking them with arrows.
- Points the robot can reach are shown clearly, while unreachable points are marked differently.
- This gives you a quick visual map of where the robot arm can or cannot work. REACH is useful for planning tasks like inspection, cleaning, painting, or maintenance because it shows whether the robot's position, size, or configuration is suitable for the job.

What is REACH?



What is Pilz Industrial Motion?

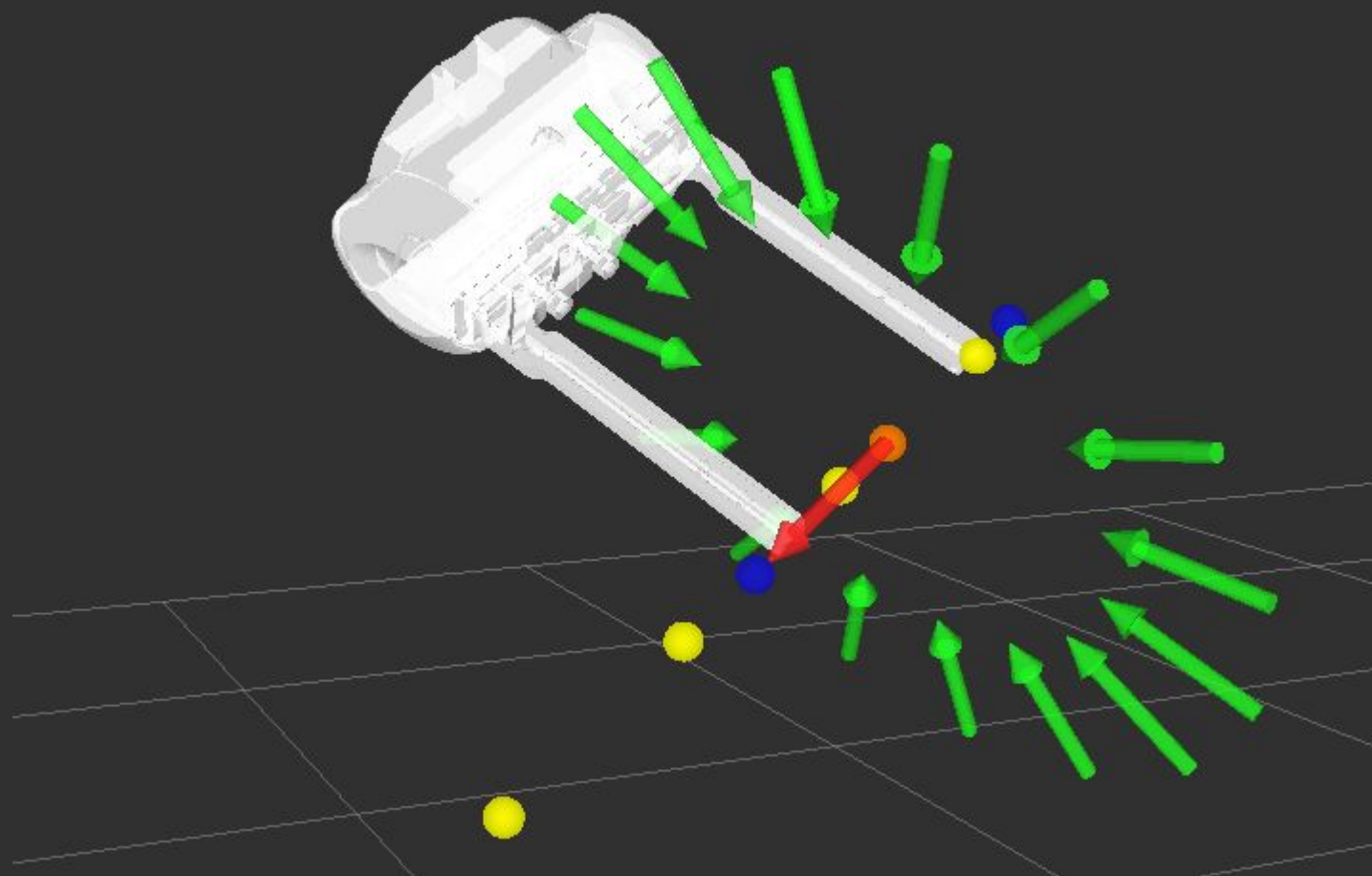
- Pilz Industrial Motion is a **tool** that makes it simple to create **Cartesian motions for industrial robots using MoveIt**.
- While most robots in ROS already support basic motion planning through RViz and MoveIt, those tools mainly focus on **free-space motion**.
- Pilz adds an easier way to plan **straight-line (LIN)**, **point-to-point (PTP)**, and **circular (CIRC) paths** in **Cartesian space**, which are commonly used in factories for tasks like welding, painting, and inspection.
- It provides a clean MoveIt interface so users can plan and run these motions without dealing with complex math.
- It also lets you blend several motions together smoothly, so the robot can perform a series of moves without stopping between them.
- This makes industrial-style robot programming easier, safer, and more predictable.

What is MoveIt Grasps?

- MoveIt Grasps is a simple tool that helps a robot figure out how to pick up basic shapes like boxes and cylinders using MoveIt's pick-and-place pipeline.
- It does not worry about advanced physics like friction; instead, it just uses geometry.
- You give it the pose of the object (where it is and how it's oriented), and it automatically creates many possible ways the gripper could approach and grasp the object.
- Then it runs inverse kinematics (IK) checks in the background to remove any grasps the robot arm cannot actually reach or perform.
- The result is a list of realistic, collision-free grasp options the robot can try.

What is MoveIt Grasps?

Grasp Quality: 92%



RViz vs MoveIt

RViz	MoveIt
A 3D visualization tool in ROS	A motion planning and manipulation framework
Shows the robot, sensors, maps, and trajectories	Computes collision-free paths and robot motions
Used to view what the robot is doing	Used to plan how the robot should move
Cannot generate motion plans	Generates motion plans using IK, planners, and controllers
Helps users interact with and inspect the robot visually	Helps robots perform tasks like grasping, picking, placing, and trajectory execution
Mainly for simulation and debugging	Used for both simulation and real robot execution
Displays results from MoveIt	Sends planned motions to RViz for display

MoveIt vs ROS 2

MoveIt	ROS 2
A motion planning and manipulation framework	A robot operating system (middleware layer)
Handles tasks like IK, collision checking, trajectory planning, and manipulation	Handles communication between nodes (topics, services, actions)
Provides high-level robotics capabilities	Provides the foundational tools robots need to run (DDS, nodes, QoS)
Works on top of ROS 2	MoveIt depends on ROS 2 to communicate and operate
Focused on arm robots, pick-and-place, and planning	Supports all types of robots: arms, drones, mobile robots, sensors, etc.
Includes planners, kinematics solvers, and perception integration	Includes communication, package management, execution, and real-time support
Helps robots decide how to move	Helps robotics software components talk to each other

Advantages of MoveIt with ROS 2

1. Real-time capability

- ROS 2 uses DDS, which supports real-time communication.
- This lets MoveIt perform faster and more predictable motion planning, especially in dynamic environments.

2. Better performance and responsiveness

- MoveIt on ROS 2 benefits from:
- Lower latency
- Multi-threading
- Improved scheduling

This means smoother and more responsive robot motions.

Advantages of MoveIt with ROS 2

3. Safer and more reliable robot operation

- ROS 2 provides:
 - Stronger QoS (Quality of Service)
 - Reliable message delivery
 - Safer node communication

This makes MoveIt more robust for industrial and commercial robots.

Advantages of MoveIt with ROS 2

4. Modern software design

- ROS 2 brings:
 - Node lifecycle management
 - Components/composable nodes
 - Cleaner APIs

This allows MoveIt to be modular, scalable, and easier to integrate into complex robotic systems.

Advantages of MoveIt with ROS 2

5. Improved multi-robot support

- ROS 2 handles multi-robot communication much better than ROS 1.
- MoveIt on ROS 2 can plan for:
 - Multiple arms
 - Mobile manipulators
 - Coordinated robots

more efficiently.

Advantages of MoveIt with ROS 2

6. Enhanced sensor integration

- ROS 2 improves data handling from:
 - Cameras
 - Depth sensors
 - LiDAR3D perception systems

MoveIt can therefore build more accurate planning scenes and adapt to changes quickly.

Advantages of MoveIt with ROS 2

7. Real-time obstacle avoidance and servoing

- ROS 1 is no longer being actively expanded.
- ROS 2 is the future of the ROS ecosystem, so MoveIt on ROS 2 ensures:
 - Long-term updates
 - Better compatibility
 - Active community support

Advantages of MoveIt with ROS 2

8. Long-term support and future-proofing

- Features like MoveIt Servo run better on ROS 2, enabling:
 - Real-time IK
 - Reactive collision avoidance
 - Teleoperation
 - Fine manipulation

Advantages of MoveIt with ROS 2

9. Commercial readiness

- ROS 2 + MoveIt powers many industrial products (e.g., MoveIt Pro). The combination is more suitable for:
 - Factories
 - Warehouses
 - Field robotics
 - Commercial deployments

Advantages of MoveIt with ROS 2

10. Better integration with modern hardware

- ROS 2 supports:
 - Real-time controllers
 - Modern middleware
 - High-speed networking

MoveIt can thus interface with new robot arms and sensors much more effectively.

Limitations of MoveIt

1. Not fully real-time by itself

- MoveIt is mainly designed for high-level planning, not hard real-time control.
- It cannot guarantee strict timing for every robot controller loop.

2. Computationally expensive for complex robots

- Planning for:
 - high-DOF arms
 - mobile manipulators
 - Humanoids

can be slow, especially in cluttered environments.

Limitations of MoveIt

3. Depends on external libraries

- Performance and reliability depend on external components like:
 - OMPL
 - FCL
 - IK solvers

If these struggle, MoveIt may struggle too.

4. Limited support for dynamic obstacles

- MoveIt handles static scenes well, but moving obstacles or fast-changing environments can reduce planning reliability.

Limitations of MoveIt

5. No built-in perception stack

- MoveIt can use sensor data but does not include:
 - image processing
 - object detection
 - grasp pose prediction

These must be added separately.

Limitations of MoveIt

6. IK solvers may fail in difficult configurations

- Default solvers (like KDL) may fail in:
 - Singularities
 - joint limits
 - complex kinematic chains

Alternative solvers (TracIK, bio_ik) help, but are not perfect.

Limitations of MoveIt

7. Smoothing and timing are not always optimal

- Generated trajectories might require:
 - extra smoothing
 - re-timing
 - post-processing

to meet industrial motion quality.

Limitations of MoveIt

8. Hard for beginners to configure

- Creating a MoveIt configuration package involves many steps:
 - URDF
 - SRDF
 - Planning groups
 - Controllers

New users may find this overwhelming.

9. Complex debugging

- Because MoveIt uses many plugins and layers, debugging failures (IK fail, controller fail, collision issues) can be difficult.

Limitations of MoveIt

10. Not ideal for very fast, reactive tasks

- Tasks like:
 - high-speed pick-and-place
 - real-time balancing
 - dynamic manipulation

may require specialized controllers or servoing systems beyond standard MoveIt planning.

Q & A





Thank
you!